

Endocrine Causes of Hypertension: Aldosterone

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INTRODUCTION

Aldosterone is a steroid hormone produced in the adrenal glands that has critical roles in blood pressure (BP), plasma volume, and potassium homeostasis. Its production is normally under feedback regulation such that hypertension, intravascular volume expansion, and hypokalemia decrease its expression. When this feedback regulation is impaired, autonomous and inappropriate aldosterone production can lead to severe hypertension that is often refractory to treatment and to hypokalemia. Autonomous aldosterone production is also associated with an increased risk of major cardiovascular events, such as myocardial infarction, congestive heart failure, atrial fibrillation, and stroke, such that the rate of these complications is substantially greater than expected solely from the associated hypertension. Recent advances in the diagnosis of aldosterone-induced hypertension have led to the recognition that primary aldosteronism is surprisingly common. Effective diagnostic strategies are available, and treatment regimens are highly efficacious.

ETIOLOGY AND PATHOGENESIS

Aldosterone is produced under normal conditions in the zona glomerulosa of the adrenal glands (Fig. 39.1). The enzyme, aldosterone synthase, is encoded by the gene *CYP11B2*, and it is generally accepted as the rate-limiting enzyme for aldosterone production. Box 39.1 summarizes factors known to stimulate or inhibit aldosterone synthesis. The physiologically most important regulators of aldosterone production are angiotensin II (Ang II), which stimulates aldosterone production through activation of the AT₁ receptor; adrenocorticotrophic hormone (ACTH), which acutely stimulates aldosterone secretion; atrial natriuretic peptide (ANP), which is inhibitory; and extracellular potassium concentration (hyperkalemia is stimulatory and hypokalemia is inhibitory).^{1,2} A high NaCl (salt) diet suppresses aldosterone production; this likely occurs through changes in ANP and renin-dependent Ang II production. Finally, there is a significant circadian variation, with serum levels greatest in the late morning with peak values about 50% greater than the average concentration.

Aldosterone mediates its effects almost exclusively through activation of the mineralocorticoid receptor (MR). MR is a classic steroid hormone receptor, which upon activation translocates to the nucleus where it functions as a transcription factor altering the expression of many genes. The endogenous glucocorticoid hormone cortisol can also activate MR. However, in most MR-expressing cells, there is concomitant expression of 11 β -hydroxysteroid dehydrogenase type 2 (11 β -HSD2), an enzyme that converts cortisol to cortisone, which does not activate MR. MR is expressed in a wide variety of tissues, including the kidney, endothelial cells, vascular smooth muscle cells, cardiomyocytes, fibroblasts, and macrophages and in the central nervous system

(CNS).³ In the kidney, MR is found in the aldosterone-sensitive distal nephron (ASDN), in glomerular mesangial cells and podocytes, and in interstitial fibroblasts.

Pathogenesis of Aldosterone-Dependent Hypertension

Aldosterone increases BP through several mechanisms. Its effects on renal NaCl retention, leading to intravascular volume expansion, are well known but are not the only mechanism. Extrarenal MR activation leads to effects on the vasculature, CNS, peripheral nervous system, and other hormones (Fig. 39.2). It also has non-hemodynamic effects, such as activation of interstitial fibroblasts, that can contribute to development of hypertension and to progression of chronic kidney disease (CKD).⁴ These multiple effects help explain why primary aldosteronism can cause refractory hypertension.

In the kidney, aldosterone stimulates renal sodium chloride retention by increasing expression of the amiloride-sensitive epithelial sodium channel (ENaC) in the collecting duct and the chloride-reabsorbing protein pendrin in the cortical collecting duct.^{5,6} Through nongenomic mechanisms, aldosterone also acutely stimulates sodium reabsorption in these segments.⁷ Furthermore, aldosterone-induced hypokalemia increases expression of the Na⁺Cl⁻ cotransport protein (NCC), the sodium-chloride cotransporter present in the distal convoluted tubule. Together, these effects on multiple Na⁺ and Cl⁻ transporters lead to NaCl retention, intravascular volume expansion, and volume-dependent hypertension.

A second renal-mediated mechanism through which aldosterone increases BP involves its generation of hypokalemia. Aldosterone increases extrarenal cellular potassium uptake by stimulating the ubiquitous Na⁺,K⁺-ATPase pump. As discussed in Chapter 10, potassium depletion raises BP through a variety of mechanisms, including increased expression of NCC in the distal convoluted tubule, induction of renal interstitial injury, and direct vascular effects.

In addition, aldosterone has multiple direct effects on the vasculature. Aldosterone increases both basal vascular tone and vascular reactivity to circulating vasoconstrictors, including norepinephrine, epinephrine, Ang II, and vasopressin.⁸ Aldosterone decreases flow-mediated vasodilation, in part by decreasing nitric oxide production as a result of decreased endothelial nitric oxide synthase expression.⁹ In addition, aldosterone leads to vascular inflammation and fibrosis, vascular stiffness, increased expression of proteins involved in atherosclerosis, and endothelial oxidative stress.¹⁰

The CNS control of BP is another mechanism through which aldosterone regulates BP. Aldosterone, through activation of CNS MR, stimulates CNS-mediated sympathetic nervous tone, which further increases BP.¹¹ CNS MR activation may also contribute to salt-sensitive hypertension and to effects of peripheral aldosterone on BP.

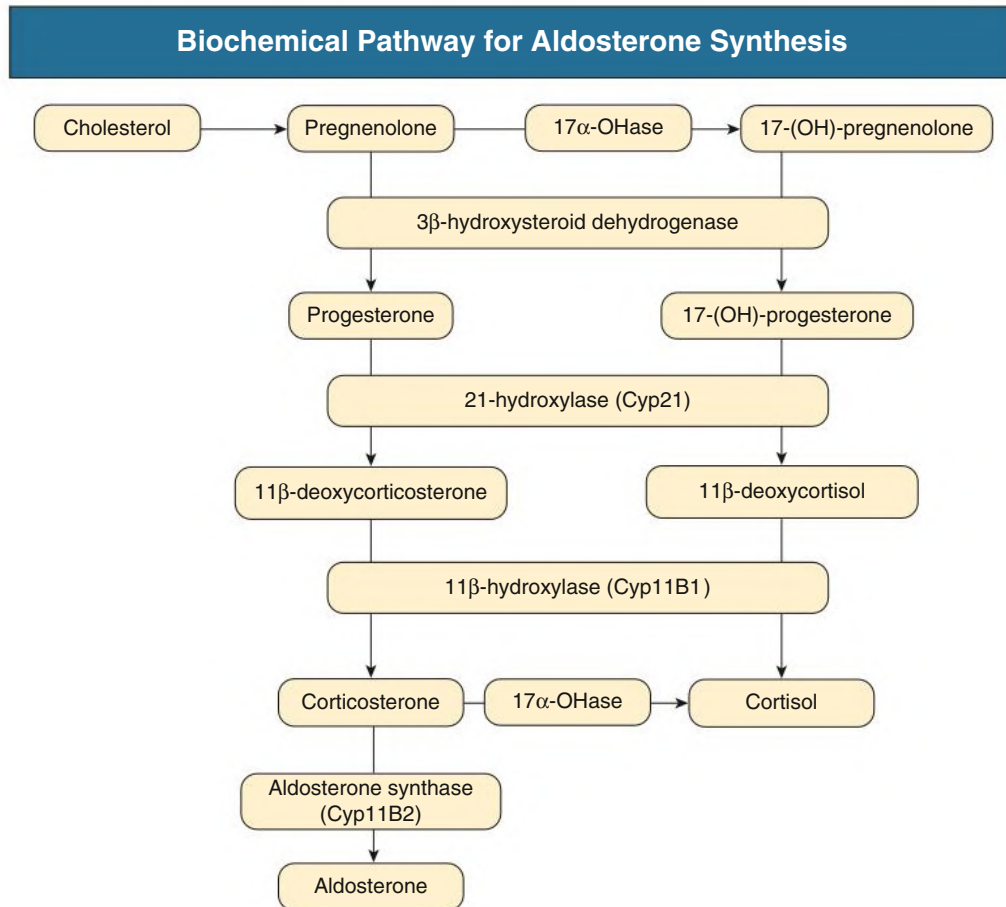


Fig. 39.1 Biochemical pathway for aldosterone synthesis.

BOX 39.1 Factors That Regulate Aldosterone Release

Stimulatory

Angiotensin II
Adrenocorticotrophic hormone^a
Potassium
Serotonin
Vasopressin
Endothelin-1
Estrogen (via GPER-1 receptor)
Parathyroid hormone
Leptin

Inhibitory

Atrial natriuretic peptide
Dopamine
Somatostatin
Estrogen (via ERβ receptor)

Both extracellular potassium and numerous hormones exert significant effects on aldosterone release. The primary effectors under most clinical circumstances are noted in italics.

^aOnly acutely.

Data from Quinn SJ, Williams GH. Regulation of aldosterone secretion. *Annu Rev Physiol.* 1988;50:409–426; and Stowasser M, Gordon RD. Primary aldosteronism: Changing definitions and new concepts of physiology and pathophysiology both inside and outside the kidney. *Physiol Rev.* 2016;96:1327–1384.

Aldosterone-Independent Mineralocorticoid Receptor Activation

MR can also be activated through mechanisms not involving aldosterone. As previously noted, cortisol, a naturally synthesized glucocorticoid, has an affinity for MR that is similar to that of aldosterone but has plasma levels about 100-fold greater than aldosterone. The enzyme 11β-hydroxysteroid dehydrogenase type 2 (11β-HSD2), which is expressed in the aldosterone-sensitive distal nephron, metabolizes cortisol to cortisone, which does not effectively bind to MR, thereby preventing glucocorticoid-dependent MR activation. Either the genetic deficiency of 11β-HSD2 or the ingestion of inhibitors of this enzyme can result in excessive activation of the MR and development of severe hypertension (see Chapters 40 and 49).¹²

A second aldosterone-independent mechanism that can activate MR involves Ang II. In the vasculature, in smooth muscle cells, Ang II acting through the angiotensin receptor blocker (ARB)-sensitive Ang II type 1 receptor activates MR through mechanisms that may involve post-translation modifications, such as phosphorylation.^{10,13}

Another aldosterone-independent mechanism involves the Rho family small guanosine triphosphatase, Rac1. Rac1 activation increases MR expression, and its expression is increased by increased dietary NaCl intake.¹⁰ This effect can lead to Rac1-dependent dietary salt-evoked paradoxical MR activation that can lead to increased renal NaCl reabsorption through reduced oxidative stress and direct regulation of Na transporters.¹⁴

Lastly, the concept of differential aldosterone sensitivity is being increasingly recognized. This concept is based on the observation that similar levels of aldosterone lead to highly different BP responses in

Effects of Aldosterone on Blood Pressure

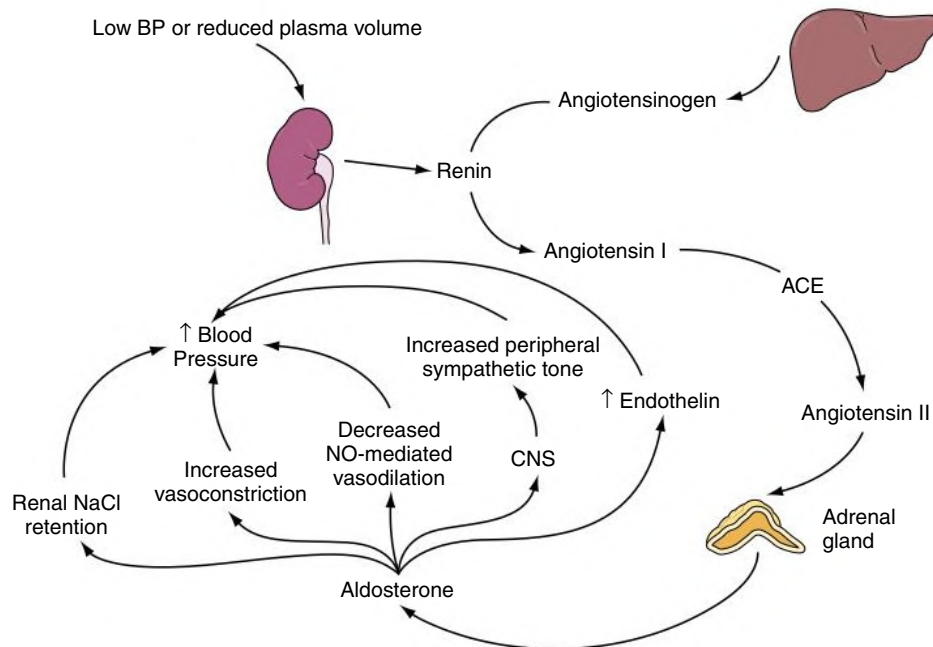


Fig. 39.2 Summary of aldosterone's effects on blood pressure (BP) regulation. *ACE*, Angiotensin-converting enzyme; *CNS*, central nervous system; *NO*, nitric oxide.

different individuals and that MR blocker (MRB) medications exert profound effects on BP in some patients, particularly those with suppressed plasma renin levels, whereas it has substantially less effect in others.¹⁵ The mechanism of this differential sensitivity is incompletely understood at present, but it is possible that differences in tissue MR expression and/or downstream signaling pathways mediate these differences.

Types of Primary Aldosteronism

Primary aldosteronism can result either from unilateral or bilateral autonomous aldosterone production (Fig. 39.3). Typically, unilateral disease results from an aldosterone-producing adenoma (APA) and bilateral disease from hyperplasia. This association is not absolute, and about 10% of patients with primary aldosteronism have either bilateral adenoma, which may be microscopic, or unilateral hyperplasia. In general, an APA causes more severe primary aldosteronism than does adrenal hyperplasia, but this distinction is not sufficient to solely guide clinical decision making.

Most APAs are unilateral and are large enough (≥ 10 mm) to be seen with thin-slice computed tomography (CT) (Fig. 39.4). However, APAs may also be microscopic, and they may be bilateral. Hyperplasia is typically bilateral but occasionally is unilateral, and it may develop asynchronously in the two adrenal glands. Because unilateral adrenal hyperplasia is potentially curable with adrenalectomy, the absence of an identifiable adrenal adenoma on CT or magnetic resonance imaging (MRI) does not exclude the possibility of surgically treatable primary aldosteronism.

Familial Hyperaldosteronism

Recognizing the presence of familial forms of primary aldosteronism is important for both case detection in affected family members and because it may result in changes in management for selected forms.

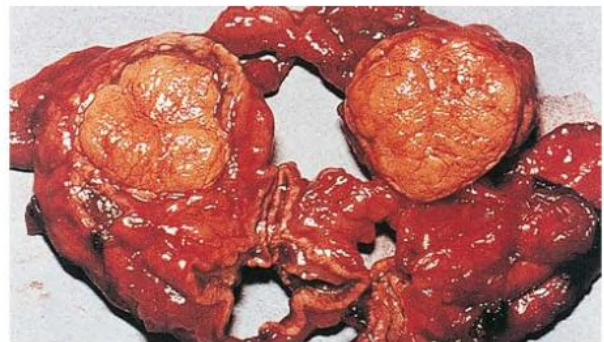


Fig. 39.3 Adrenal Adenoma. An aldosterone-producing adrenal adenoma with typical cholesterol-rich yellow appearance.

Although identification of familial hyperaldosteronism (FH) was previously based on an arbitrary definition of the presence of two or more family members, improved understanding of the genetic causes has led to a classification system based on genomic DNA sequencing findings. Table 39.1 summarizes the different forms of FH. In some cases, the clinical presentation suggests the presence of a familial etiology, and in some the presentation is indistinguishable from sporadic forms. Except for familial hyperaldosteronism type I, identification of these familial forms, beyond allowing specific genetic testing of family members of affected individuals, does not alter treatment decisions at this time.

The first identified was *familial hypertension type I* (FH-I), which is also known as *glucocorticoid-remediable aldosteronism* (GRA). In FH-I there is crossover between the *CYP11B1* and *CYP11B2* genes. This leads to a chimeric aldosterone synthase gene whose expression is regulated by ACTH, leading to excessive aldosterone release.^{16,17} FH-I is transmitted in an autosomal dominant pattern and should be considered in children or young adults with refractory hypertension or

who have a family history of hypertension at a young age or history of premature hemorrhagic stroke.¹⁸ If suspected, genetic testing is the preferred diagnostic approach because this offers improved sensitivity and specificity over measurement of steroid metabolites or dexamethasone suppression testing.¹⁸ When FH-I is found, administering corticosteroids at the lowest dose necessary to suppress ACTH release often dramatically improves BP control (see Chapter 49). Concomitant MRB and/or anti-ENaC therapy may be necessary to avoid high doses of glucocorticoids with their associated side effects.

Familial hyperaldosteronism type II. In the past several years, there has been a fundamental change in our understanding of FH

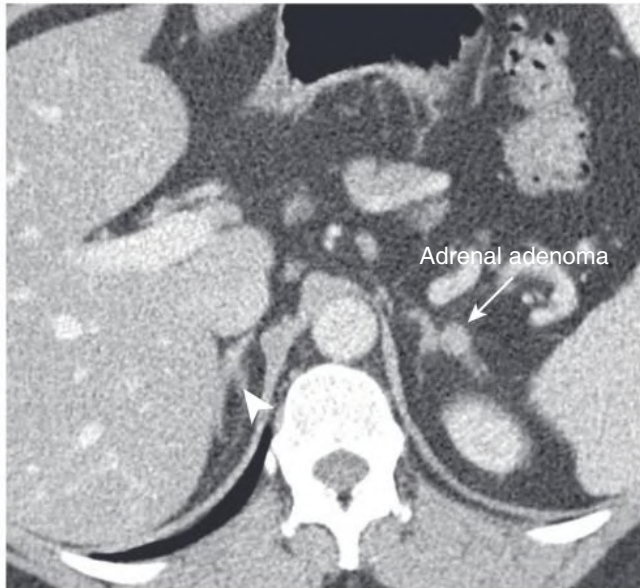


Fig. 39.4 Adrenal Adenoma by Computed Tomography Scan. A normal linear image of the right adrenal gland (arrowhead) and expansion of the left adrenal with aldosterone-producing adenoma (~1 cm) (arrow).

type II (FH-II). Previously, FH-II was based on the presence of at least two family members with primary aldosteronism after excluding a diagnosis of FH-I. This definition almost certainly overestimated the frequency of FH because of misclassification of sporadic primary aldosteronism in multiple family members as FH-II. FH-II is now known to result from germline mutations in the gene, *CLCN2*, which encodes the chloride channel, *ClC-2*.^{19,20} These mutations lead to gain-of-function, increasing cellular Cl^- exit, which causes membrane depolarization. Membrane depolarization increases Ca^{2+} entry through voltage-sensitive calcium channels. Increases in intracellular Ca^{2+} lead to cellular proliferation and increased *CYP11B2* expression, which increases aldosterone production and secretion. FH-II appears to present in childhood with severe primary aldosteronism.

Familial hyperaldosteronism type III. FH type III (FH-III) is the result of germline mutations in the *KCNJ5* gene.^{21–23} The *KCNJ5* gene encodes a K^+ -channel that is a major determinant of adrenal cortical cell membrane voltage. Mutations observed in FH-III result in loss of K^+ selectivity, leading to Na^+ influx and membrane depolarization. Consequently, there is voltage-gated Ca^{2+} channel activation, increased cytosolic calcium, and stimulation of aldosterone production and adrenal cell proliferation. Although initial studies characterized FH-3 as having severe, often childhood-onset of hypertension, subsequent studies suggest the presentation is more variable.²⁴

Familial hyperaldosteronism type IV. FH type IV (FH-IV) results from mutations in the *CACNA1H* gene, which encodes for a T-type calcium channel ($\text{Ca}_v3.2$).²⁵ These mutations appear to lead to increased Ca^{2+} entry, increased cytosolic calcium, and stimulation of aldosterone production and adrenal cell proliferation. Clinical penetrance of the mutation is incomplete.

Sporadic Forms

Our understanding of the factors that lead to sporadic forms of primary aldosteronism has advanced substantially. This is true for both unilateral autonomous aldosterone production associated with an APA and for bilateral autonomous aldosterone production. DNA sequencing,

TABLE 39.1 Familial Hyperaldosteronism (FH)

Form	Affected Gene	Pathogenic Effect	Classic Presentation	Targeted Treatment
FH-I	Crossover of <i>CYP11B1</i> promoter with <i>CYP11B2</i> coding sequence	Aldosterone synthase expression under control of ACTH-sensitive promoter	Severe hyperaldosteronism in childhood or young adults with autosomal dominant inheritance	Low-dose glucocorticoids, to suppress endogenous ACTH production. May need concomitant ARB and/or ENaC inhibitor therapy.
FH-II	<i>CLCN2</i>	Gain-of-function mutation in the Cl^- channel <i>ClC-2</i> , leading to increased Cl^- efflux and subsequent membrane depolarization Membrane depolarization increases Ca^{2+} entry, which leads to hyperplasia and increased <i>CYP11B2</i> expression	Early-onset hypertension	None available
FH-III	<i>KCNJ5</i>	Loss of Na^+ selectivity of K^+ channel, GIRK4, leading to membrane depolarization, which increases Ca^{2+} entry and subsequent hyperplasia and increased <i>CYP11B2</i> expression	Severe early-onset hypertension associated with profound hypokalemia	None currently available at present Macrolide antibiotics are effective in vitro, but in vivo effects have not been well characterized
FH-IV	<i>CACNA1H</i>	Mutation in the pore-forming $\alpha 1$ subunit of the voltage-sensitive calcium channel <i>Ca_v3.2</i> , leading to increased Ca^{2+} entry and subsequent hyperplasia and increased <i>CYP11B2</i> expression	Early-onset primary aldosteronism May be associated with developmental delay or attention deficit disorder	None currently available

ACTH, Adrenocorticotrophic hormone; ARB, angiotensin receptor blocker; ENaC, epithelial sodium channel.

either “classic” Sanger sequencing, or immunohistochemistry-guided next-generation sequencing, shows that acquired somatic mutations lead to many cases of sporadic primary aldosteronism.^{19,26} Unilateral autonomous aldosterone production associated with an APA is frequently associated with *KCNJ5* mutation in the APA and is found in as many as 80% of APA, with a higher frequency in Asian populations.^{19,27} In African Americans, somatic *CACNA1D* mutations appear to be the most frequently acquired mutation.²⁸ Other genes in which somatic mutations have been found frequently include *ATP1A1* and *ATP2B3*. Less frequently, mutations in *CTNNB1* and *PRKACA* have been identified.^{19,27}

Recent advances have also been made in understanding the pathogenesis of primary aldosteronism from bilateral adrenal hyperplasia. Activating autoantibodies directed against the angiotensin II type 1 (AT1) receptor may activate the AT1 receptor, both directly stimulating adrenal aldosterone secretion and contributing to the hypertension through an aldosterone-independent mechanism involving AT1 receptor activation in vascular tissue.^{29,30} To date, the presence of these antibodies has been examined in only a limited number of patients. Additional studies are needed to examine how widespread this mechanism is and what proportion of cases of bilateral adrenal hyperplasia results from this mechanism.

EPIDEMIOLOGY

The apparent incidence of primary aldosteronism varies with the patient population and the diagnostic criteria used. Early studies, which only recognized severe cases, suggested that primary aldosteronism was rare, with an incidence of less than 1% to 2%. More sensitive diagnostic criteria have now led to the recognition that primary aldosteronism is relatively common.³¹ Patients with treatment-resistant hypertension—inadequately controlled hypertension despite treatment with three medications at appropriate dosages, including a diuretic—are quite likely to have primary aldosteronism, with rates typically of 20% to 40% and as high as 67% in some studies.³²

Primary aldosteronism causes adverse effects that are independent of hypertension and hypokalemia. This includes an increased risk of cardiac arrhythmias, both atrial and ventricular, cerebrovascular events, and coronary heart disease.³³ Diagnosing and treating the primary aldosteronism, whether with adrenalectomy or with MRBs, as clinically appropriate, substantially reduces, and may even fully correct, this increased risk.^{34,35} Finally, primary aldosteronism is associated with a variety of neuropsychiatric disorders, ranging from depression and anxiety to lethargy, fatigue, and difficulties with concentration. Appropriate treatment of the primary aldosteronism may improve these symptoms.²⁴

CLINICAL MANIFESTATIONS

Identifying patients with primary aldosteronism purely on clinical characteristics is difficult. Some patients with primary aldosteronism have features suggestive of secondary hypertension, such as early onset of hypertension or the need for multiple drugs for BP control. Others present with either frank or easily provoked hypokalemia. In hypertensive patients with atrial fibrillation without a known cause of atrial fibrillation, primary aldosteronism may be present in as many as 42%.³⁶ Many, however, have no distinguishing characteristics that differentiate them from individuals with essential hypertension. The incidence of primary aldosteronism in a hypertensive population parallels the severity of hypertension, although 1% to 2%, and possibly as many as 13%, of normotensive individuals show evidence of autonomous aldosterone production (Fig. 39.5).^{31,37}

EVALUATION AND DIFFERENTIAL DIAGNOSIS

Who to Evaluate for Primary Aldosteronism

We recommend a liberal policy for testing patients for primary aldosteronism. This involves testing patients who have early-onset hypertension, baseline or easily provoked hypokalemia, or hypertension not effectively controlled with routine antihypertensive therapy. Some evidence suggests patients with hypertension and atrial fibrillation should be screened for primary aldosteronism.³⁶ Patients with hypertension and a known adrenal adenoma should also undergo testing. Finally, hypertensive patients with a family history of primary aldosteronism should be tested. It is important to recognize that the absence of hypokalemia or metabolic alkalosis does not exclude the diagnosis of primary aldosteronism.

Evaluation of Suspected Primary Aldosteronism

Evaluation of patients with suspected primary aldosteronism is directed at identifying those who have autonomous aldosterone release, followed by determining whether treatment should be based on a pharmacologic or a surgical approach (Fig. 39.6).

The diagnosis of primary aldosteronism requires evidence of autonomous aldosterone production. As previously detailed, the major factors that stimulate aldosterone production involve Ang II-dependent and hyperkalemia-dependent stimulation of the adrenal gland. Thus, autonomous aldosterone production is identified by inappropriate aldosterone production in view of Ang II and serum/plasma K^+ . Hyperkalemia, if present, should be corrected before screening to eliminate this possible stimulus for aldosterone production. Because Ang II cannot be assayed with routine clinical tests, plasma renin is used as a surrogate. A random blood sample is used to measure plasma aldosterone and renin. This test should be obtained in early morning because of diurnal variations in plasma aldosterone and plasma renin activity. The question as to whether to hold medications that can alter the renin-angiotensin-aldosterone system and, if so, for how long before testing is complex and is discussed in detail later. If the aldosterone/renin ratio (ARR) is elevated, this suggests there is significant Ang II-independent aldosterone release, providing presumptive evidence of autonomous aldosterone production consistent with primary aldosteronism.

Incidence of Primary Hyperaldosteronism in Patients With Differing Degrees of Hypertension

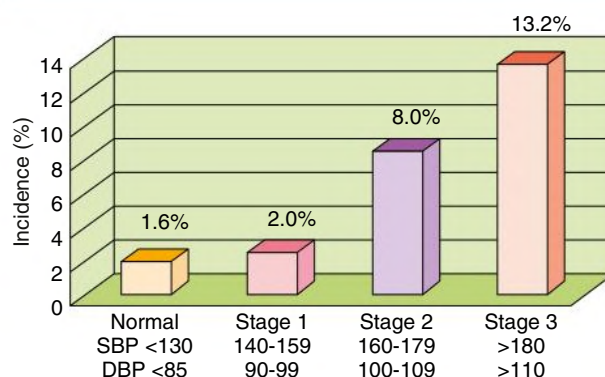


Fig. 39.5 Incidence of primary aldosteronism in patients with differing degrees of hypertension. *DBP*, Diastolic blood pressure; *SBP*, systolic blood pressure. (From Mosso L, Carvajal C, Gonzalez A, Barraza A, Avila F, Montero J, et al. Primary aldosteronism and hypertensive disease. *Hypertension*. 2003;42:161–165.)

TABLE 39.2 Confirmatory Testing for Primary Aldosteronism

Test	Method	Evaluation	Limitations
Oral salt loading	Oral NaCl intake >200 mmol/d for 3 days, with oral KCl as needed to prevent hypokalemia, with subsequent 24-hr urine aldosterone measurement	Urine aldosterone <10 µg/d, diagnosis unlikely; >12 µg/d, primary aldosteronism likely	Avoid if severe uncontrolled hypertension, CKD, CHF, cardiac arrhythmias, or severe hypokalemia
Saline infusion test	Patient in recumbent position for 1 hr before testing and then throughout entire test Begin test between 8 and 9:30 AM; measure plasma aldosterone, plasma renin activity, cortisol, and potassium at beginning of test and then after infusing 2 L NS IV over 4 hr	Plasma aldosterone at end of infusion <5 ng/dL, primary aldosteronism unlikely; >10 ng/dL, diagnosis likely; 5–10 ng/dL, indeterminate	Avoid if severe uncontrolled hypertension, CKD, CHF, cardiac arrhythmias, or severe hypokalemia
Fludrocortisone suppression test	Oral fludrocortisone, 0.1 mg every 6 hr for 4 d, plus oral NaCl, 30 mmol 3×/d, and high-salt diet combined with sufficient KCl to avoid hypokalemia	Upright plasma aldosterone on day 4 >6 ng/dL and plasma renin activity <1 ng/mL/hr, primary aldosteronism likely	Requires very close follow-up to monitor blood pressure and potassium
Captopril challenge test	Plasma aldosterone and plasma renin activity obtained immediately before oral captopril, 25–50 mg, is given and then 1–2 hr afterward, with patient seated throughout test	Plasma aldosterone decrease >30%, primary aldosteronism unlikely	Probably more false-positive and false-negative results than other tests

CHF, congestive heart failure; CKD, chronic kidney disease; IV, intravenously; NS, normal saline.

Although an elevated aldosterone in combination with an elevated ARR strongly suggests primary aldosteronism, the ARR may be elevated even with suppressed plasma aldosterone production if plasma renin activity (or concentration) is substantially suppressed. In this event, the elevated ARR may represent a false-positive screening test. We recommend that a combination of an elevated ARR and a “non-suppressed” aldosterone level should be used for case-screening for primary aldosteronism. The minimal plasma aldosterone level that may be associated with aldosteronism is unclear. We recommend using a minimal value of 8 ng/dL to make a diagnosis of primary aldosteronism.³⁸ Others recommend using a cutoff of 15 or even 20 ng/dL.^{39,40} We recommend the lower cutoff because about 40% of patients with primary aldosteronism have plasma aldosterone levels between 9 and 16 ng/dL,^{37,41} and about 20% of those with unilateral autonomous adrenal aldosterone production have levels less than 15 ng/dL.⁴¹

Effect of Medications on Screening for Primary Aldosteronism

Many medicines commonly used for the treatment of hypertension can alter either plasma aldosterone or renin release and thereby alter the ARR.^{40,42} β-Adrenergic receptor antagonists (β-blockers) suppress renin but generally do not cause the nearly complete renin suppression typically seen in primary aldosteronism. Angiotensin-converting enzyme (ACE) inhibitors (ACE-I) and ARBs, along with diuretics, can increase renin release in normal individuals, which theoretically might decrease the sensitivity of ARR measurement. However, the effect of ACE-Is and ARBs to increase renin release can also be an advantage. In patients using either an ACE-I or an ARB, a suppressed renin in combination with a nonsuppressed aldosterone (>8 ng/dL) is highly specific for primary aldosteronism. MRBs such as spironolactone and eplerenone can elevate the plasma renin activity and impair the sensitivity of testing. Direct renin inhibitors have been reported to both increase and decrease plasma renin activity in patients who presumably have essential hypertension and to decrease plasma aldosterone. Their effect on plasma renin activity and ARR in patients with primary aldosteronism is presently not known.

When to Hold Medications That Affect the RAS System Before Screening

The decision whether to hold medications that affect the renin-angiotensin-aldosterone system requires the clinician to balance

competing priorities. Holding these medications, which includes diuretics, ACE-Is, ARBs, direct renin inhibitors, and β-blockers, improves the diagnostic accuracy of the plasma aldosterone and plasma renin activity. However, in the patient with poorly controlled hypertension, discontinuing these medications and attempting to control the blood pressure solely with calcium channel blockers, α-adrenergic blockers, and direct vasodilators can be problematic. In one study, as many as 38% of patients with difficult-to-control hypertension experienced one or more related adverse effects.⁴³ Moreover, the amount of time drugs should be stopped to completely reverse their effects on the adrenal gland is not clear.

The observation that patients with refractory hypertension who do not have primary aldosteronism respond similarly well to MRB therapy as those who do^{44,45} suggests that accurate diagnosis may only be needed to guide whether to pursue evaluation of unilateral autonomous aldosterone production. If medications are to be withheld, it should be done carefully using standardized protocols with extensive safety monitoring.

In patients being evaluated because they have an adrenal adenoma and who do not have resistant hypertension or substantial hypokalemia, the risk/benefit balance may differ. In this latter group, the clinical difficulty and the risks to the patient from changing medications are less substantial, and the utility of accurately identifying clinical milder forms of primary aldosteronism may be greater.

Confirmatory Testing

If the diagnosis of primary aldosteronism is in doubt, several confirmatory tests can be used. Table 39.2 summarizes the various confirmatory testing methods in routine use. Recent studies suggest that confirmatory testing may not be needed in patients with spontaneous hypokalemia, plasma aldosterone greater than 20 ng/dL, and completely suppressed renin⁴⁶ or in patients with plasma aldosterone greater than 30 ng/dL with a suppressed renin.⁴⁷ Both the risks and the benefits of confirmatory testing must be considered.³⁸ No confirmatory test has 100% sensitivity or specificity. Moreover, confirmatory testing has complicating issues associated with its performance. Salt loading, if used for testing, can raise BP and worsen hypokalemia and is problematic in the patient who already has poorly controlled hypertension. Testing also requires the patient not use any medications that interact with the RAS, which means discontinuing β-blockers, ACE-Is,

ARBs, MRBs, direct renin inhibitors, and diuretics; this can lead to significant worsening of hypertension control.³⁸ How long these medications should be held is not clear. Acute effects will dissipate within days as the medications are metabolized, but chronic effects on expression of proteins involved in adrenal aldosterone metabolism may persist longer.

Confirmatory testing should be considered if the diagnosis is in doubt and if it will substantially alter management. In general, patients with resistant hypertension respond to MRBs with significant BP improvement regardless of whether they have primary aldosteronism.^{44,45} Because of this, confirmatory testing may not alter the decision as to whether to use MRB therapy in this population. We generally reserve confirmatory tests for patients with an adrenal adenoma found for unrelated reasons who have mild hypertension and are under evaluation for possible adrenalectomy. In this latter patient population, the benefits of MRB therapy are less clear, and the clinical risks associated with holding RAS blockers are substantially less.

Unilateral Versus Bilateral Differentiation

Selection of Patients for Further Testing

Once primary aldosteronism is diagnosed, the clinician should consider the possibility that the patient has unilateral aldosterone release, in which case adrenalectomy may be curative. However, this step is important only if the patient is both an appropriate surgical candidate and the patient desires possible surgical intervention. If either condition is not met, then further evaluation is not necessary, and the patient can be treated with MRB therapy.

Imaging Approaches

Adrenal imaging is typically the next step in evaluation. Unilateral autonomous aldosterone production is often the result of an aldosterone-producing adenoma (see Fig. 39.3), which is frequently sufficiently large to be identified by thin-slice (1 mm) CT (see Fig. 39.4). However, in some cases, microadenoma that are not detectable with current imaging approaches are the cause of unilateral autonomous aldosterone release. Bilateral autonomous aldosterone production is typically not associated with the presence of a detectable functional adenoma, but nonfunctional adrenal adenomas are frequent, and their incidence increases with age. Consequently, the presence of an adenoma on imaging studies does not necessarily indicate unilateral disease, and its absence does not necessarily indicate bilateral aldosterone production. Overall, an adrenal adenoma in a patient with primary aldosteronism has an approximately 70% to 80% chance that it is functional (i.e., an APA is present) and a 20% to 30% chance that it is a nonfunctional adenoma.⁴⁸ In patients younger than 40 years old, the likelihood of a nonfunctional adenoma is less, and the likelihood of an APA is greater. The absence of a detectable adenoma indicates bilateral aldosterone production in 70% to 80% of cases, but there is a 20% to 30% chance of unilateral aldosterone production.

Adrenal Vein Sampling

Adrenal vein sampling is the definitive test for determining whether autonomous aldosterone release is unilateral or bilateral. This is a technically difficult procedure and should be performed in a specialized center experienced with it. When interpreting adrenal vein sampling results, the first step is to confirm sampling of both the right and left adrenal veins. The right adrenal vein is short and may have a common insertion into the inferior vena cava (IVC) as the inferior accessory hepatic vein, making successful cannulation difficult. Accordingly, biochemical confirmation of successful adrenal vein sampling should be employed. This is done by measuring the concentration of a hormone produced in the adrenal gland and showing that the observed

TABLE 39.3 Formulas Used to Interpret Adrenal Vein Sampling Results

Name	Formula	Comment
Selectivity index (SI)	$SI_X = [Cortisol]_X / [Cortisol]_{IVC}$	Metanephine or androstenedione measurements may be substituted for cortisol.
Lateralization index (LI)	$LI = \frac{[Aldo_A] / [Cortisol_A]}{[Aldo_B] / [Cortisol_B]}$	Calculate using the adrenal vein sample with the higher aldosterone concentration as sample A and the contralateral samples as sample B.

concentration is significantly greater than in the IVC. To quantitatively evaluate whether the adrenal veins were successfully cannulated, a selectivity index is calculated from the adrenal vein and IVC cortisol concentrations (Table 39.3). The higher the selectivity index, particularly if it is greater than 2 in unstimulated and greater than 5 in stimulated samples, the greater the confidence that the sample accurately reflects adrenal origin.⁴⁹ Cortisol is the most frequently measured analyte, but metanephine or androstenedione can also be used.^{50,51} Stimulation of cortisol production with cosyntropin can facilitate confirmation of adrenal vein cannulation by increasing adrenal cortisol production, which increases adrenal vein cortisol levels acutely with minimal effects on IVC cortisol. However, cosyntropin can also stimulate aldosterone secretion, which can lead to differing conclusions as to whether aldosterone production is unilateral or bilateral in as many as 25% of cases.⁵²

The lateralization index (LI) is then determined to assess aldosterone lateralization. After confirming successful adrenal vein sampling, the ratio of adrenal vein aldosterone to adrenal vein cortisol is calculated for each adrenal vein (see Table 39.2). The LI is calculated by dividing the greater adrenal vein ratio by the lesser adrenal vein ratio. An LI greater than 2.0 supports a diagnosis of an aldosterone-producing adenoma, but this cutoff is not 100% specific. Our practice is to use an LI greater than 2.0 to indicate an aldosterone-producing adenoma if this is concordant with imaging findings. If the LI finding is discordant with imaging, we consider a LI between 2 and 4 an indeterminant finding. The importance of interpreting adrenal vein sampling in context of the clinical situation is exemplified by the finding that adrenal vein sampling performed twice within 5 minutes had a discordance rate of 13% for the selectivity index and 10% for the LI.⁵³

Other Testing Options

Many other diagnostic tests have been suggested in the evaluation of primary aldosteronism to differentiate unilateral from bilateral disease, but none has widespread acceptance. Noninvasive physiologic tests, such as saline suppression test and postural stimulation tests, rely on the association that APA-mediated aldosterone release is typically not stimulated by Ang II, whereas in hyperplastic lesions, Ang II-dependent stimulation is generally retained. All medicines that affect the RAS, including diuretics, β -blockers, ACE-Is, and ARBs, must be stopped before testing, but the difficulty in controlling BP in these patients without these medicines often limits the utility of this test. In addition, the association of Ang II-stimulated aldosterone production is discordant with whether aldosterone release is unilateral or bilateral in as many as 20% of cases, leaving this test result insufficiently accurate to determine whether surgical therapy is appropriate in most

cases. Its main role may be in the evaluation of the patient with an adrenal adenoma and unsuccessful adrenal vein sampling.⁵⁴

Aldosterone Breakthrough

Approximately 30% to 40% of patients treated with an ACE-I or ARB respond with an initial decrease in the plasma aldosterone levels, which is then followed by a return of their aldosterone level to baseline, or even above baseline, levels. This is termed *aldosterone breakthrough*.⁵⁵ Mechanisms that have been proposed to explain this include ACE-independent Ang II generation, Ang II-independent stimulation of aldosterone generation, possibly via either ACTH or ANP, and alterations in endothelial nitric oxide formation.^{55,56} Aldosterone breakthrough likely contributes to the development of resistant hypertension in patients being treated with an ACE-I or ARB and likely contributes to progressive kidney disease in CKD.

Metabolic Syndrome

The metabolic syndrome is associated with elevated plasma aldosterone levels,⁵⁷ which may contribute to the associated, and often resistant, hypertension. However, it does not appear to be associated with changes in plasma renin activity.⁵⁷ The mechanism of the increased plasma aldosterone may involve adipocyte production of cytokines that stimulate adrenal sensitivity to Ang II.⁵⁸ It may also involve fructose, a common pathogenic factor in the metabolic syndrome, stimulating vasopressin release, which stimulates ACTH release⁵⁹ and thereby increases aldosterone production. Vasopressin may also directly increase adrenal aldosterone secretion.⁶⁰

NATURAL HISTORY

The natural history of untreated primary aldosteronism leads to a variety of significant complications. Clearly, it is a major cause of worsening hypertension, and it also increases the incidence of hypokalemia, with its adverse chronic effects. In addition, it leads to a wide variety of end-organ complications. These end-organ effects include increased risks of coronary heart disease, atherosclerotic peripheral vascular disease, cardiac arrhythmias, left ventricular hypertrophy, low bone mineral density, and development of CKD.^{33,61} The incidence of these complications appears to be twofold greater than in patients with essential hypertension of the same magnitude. These effects appear to be related to excessive activation of mineralocorticoid receptors in these end-organ tissues and therefore reflect a direct, and blood pressure-independent, disease association. Effective treatment of primary aldosteronism appears to substantially improve these risks. Therefore, treatment either with MRBs or with adrenalectomy is recommended for all patients with primary aldosteronism.

TREATMENT

Adrenalectomy

Patients with unilateral autonomous aldosterone production, typically from an APA, who are acceptable surgical candidates, can have a dramatic response to adrenalectomy. Patients with an APA who undergo adrenalectomy have a greater than 95% likelihood of biochemical cure and a 30% to 60% chance of hypertension cure.^{62,63} Patients most likely to have their hypertension cured are female, younger than 50 years, and have shorter duration of hypertension, lower body mass index, and few family members with primary hypertension. Those not cured have a greater than 95% chance of BP improvement.^{62,63} The lack of complete cure may reflect microvascular disease that developed because of the poorly controlled hypertension and/or hypokalemia. Adrenalectomy performed using a laparoscopic approach is associated with a relatively short hospital stay, low postoperative morbidity, and a

rather quick return to normal health, at least compared with a nonlaparoscopic surgical approach. Adrenalectomy should not be performed in patients with bilateral autonomous aldosterone production because unilateral adrenalectomy is not curative and bilateral adrenalectomy is associated with unacceptable long-term complications because of the induction of deficiency of other adrenal hormones, such as glucocorticoids and catecholamines.

Mineralocorticoid Receptor Blocker Therapy

Patients who are not candidates for surgical adrenalectomy or who have bilateral aldosterone production should be treated with MRBs. Current MRB treatment options are spironolactone and eplerenone. Many patients will find that their responses are delayed, and monthly dose adjustments are often needed until an optimal dose is identified. It is possible that some of the effects of MRB therapy may also be mediated through effects on pannexin.⁶⁴

The choice of which MRB to use involves consideration of effectiveness and side effects. Importantly, the treatment efficacy of these two agents is quite different. Spironolactone is more effective than eplerenone at improving BP, yielding almost twice as great a BP reduction and a substantially greater proportion of patients experiencing a significant BP improvement.⁶⁵ Spironolactone also has off-target effects on sex-steroid hormone receptors, which can lead to breast enlargement (gynecomastia) in men and breast tenderness (gynecomdynia) and menstrual irregularities in women. These side effects are dose-dependent and are typically mild or absent at doses of 100 mg per day or less. Eplerenone is more selective than spironolactone for the mineralocorticoid receptor, and the risk of these side effects is less, but its efficacy is also less.

Based on its greater treatment efficacy, we recommend spironolactone as the preferred initial agent for medical treatment of primary aldosteronism. Treatment with an MRB usually results in dramatically improved BP control. Many patients will respond to a low dose of spironolactone (25–50 mg daily) as initial therapy. Both systolic BP and diastolic BP often decrease by about 25 mm Hg over a few weeks to months. Dosage of the MRBs can be increased as necessary but generally should not be changed more than every 2 to 4 weeks. If present, hypokalemia and metabolic alkalosis typically improve. Potassium supplements can often be stopped when MR therapy is initiated; if not, the dose can often be tapered rapidly. Because there is significant interpatient variability in the serum K⁺ response after initiating MRB therapy, close follow-up is needed.

With time, BP can be controlled in many patients with an MRB and a single alternative agent. We typically use ACE-Is or ARBs because renin activity, which is suppressed initially, typically increases after starting MRB therapy. Synergistic use of an ACE-I or ARB may prevent development of a renin-stimulated, Ang II-dependent component of hypertension. However, many antihypertensive combinations can be used successfully with the MRB.

Many of the side effects of MRBs can be easily managed. Muscle cramps occur with both spironolactone and eplerenone. Changing from one MRB to the other often allows resolution of the muscle cramps. MRBs can cause neuropsychiatric side effects that are often nonspecific and can include symptoms of feeling out of touch with reality, difficulty thinking, and decreased attention and performance in various memory tasks.^{65,66} In general, these effects are temporary and resolve within a few weeks.

Other MRBs are in development and may become available for use soon. Finerenone is a novel, selective, nonsteroidal MRB that may have less effect on serum potassium and slows the progression of diabetic nephropathy.⁶⁷ This may make this medication useful in selected patients. Multiple other nonsteroidal MRBs are also in development.⁶⁸

Plasma Renin Activity Predicts Risk of Cardiovascular Events During Mineralocorticoid Receptor Blocker Therapy

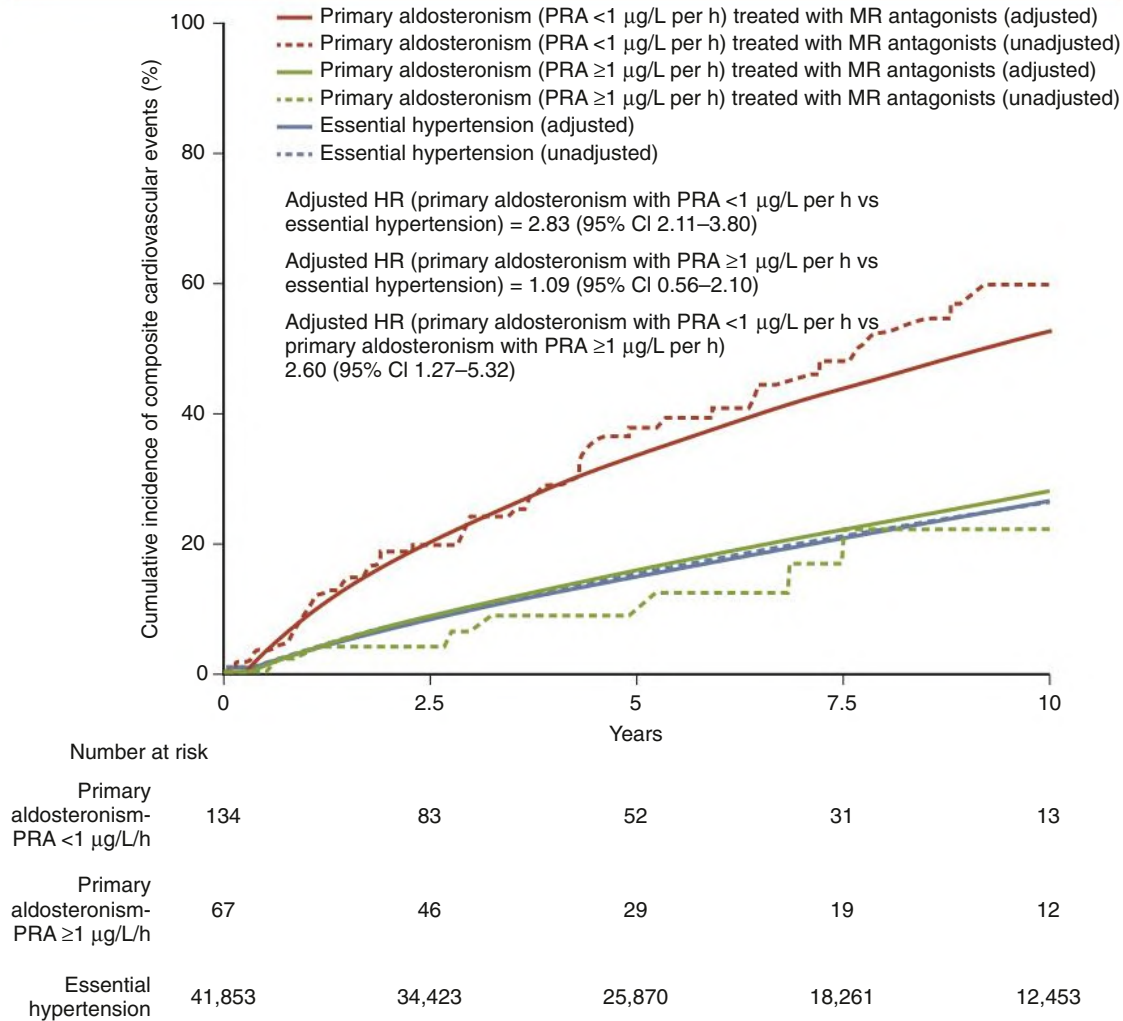


Fig. 39.7 Plasma Renin Activity Predicts Risk of Cardiovascular Events During Mineralocorticoid Receptor Blocker (MRB) Therapy of Primary Aldosteronism. Patients with primary aldosteronism treated with MRBs that have plasma renin activity of at least 1 ng/mL/hr have a composite risk of cardiovascular events not significantly different from matched patients with essential hypertension. Those patients with plasma renin activity less than 1 ng/mL/hr have a significantly increased hazard ratio (HR) for composite cardiovascular events 2.83-fold greater. *CI*, Confidence interval; *PRA*, plasma renin activity; *MR*, mineralocorticoid receptor. (From Hundemer GL, Curhan GC, Yozamp N, Wang M, Vaidya A. Cardiometabolic outcomes and mortality in medically treated primary aldosteronism: a retrospective cohort study. *Lancet Diabetes Endocrinol*. 2018;6:51–59.)

Monitoring Mineralocorticoid Receptor Blocker Therapy With Plasma Renin Measurements

Primary aldosteronism is associated with end-organ complications, particularly affecting the cardiovascular system, whose frequency is greater than that expected from the associated hypertension. This suggests that treatment of primary aldosteronism directed solely at controlling BP may not fully reverse the adverse consequences of primary aldosteronism. Assessment of the plasma renin activity response during therapy may be helpful (Fig. 39.7). Recent studies show that MRB therapy that results in reversal of the plasma renin suppression, to greater than 1 ng/mL/hr, is associated with major cardiovascular events, mortality, and atrial fibrillation rates similar to matched patients with essential hypertension.^{34,35} In contrast, in those treated with MRB therapy who continue to have suppressed plasma renin

activity, less than 1 ng/mL/hr, these rates remain approximately two-fold elevated.^{34,35} Although alternative explanations other than therapy effectiveness are possible, we recommend using a plasma renin-guided therapeutic approach, adjusting the MRB dose on a 1- to 2-month basis targeting a plasma renin activity greater than 1 ng/mL/hr.

Choice of Adrenalectomy Versus MRB Therapy

The choice of adrenalectomy versus MRB therapy for the patient with unilateral autonomous aldosterone production is often not easy. Adrenalectomy is associated with a greater than 95% biochemical cure, and with clinical cure (normotension and normokalemia without medications) of 30% to 60%. Studies comparing outcomes in patients with unilateral primary aldosteronism treated with adrenalectomy versus medical therapy have suggested benefits from adrenalectomy in terms of lower all-cause mortality, reduced risk of stroke and atrial fibrillation,

and regression of left ventricular hypertrophy in patients treated with adrenalectomy rather than MRB therapy.^{19,69} However, whether the patients treated with MRB therapy had normalization of the plasma renin is unknown, and if they did not, whether renin-guided medical therapy would have resulted in equivalent outcomes as adrenalectomy is also unknown. Finally, many patients are elderly, have significant comorbid medical conditions that make adrenalectomy relatively contraindicated, or prefer not to undergo elective surgery.

Nonmineralocorticoid Receptor Blocker Therapy

In occasional patients, high doses of MRBs will not be sufficient to block the effects of aldosterone on BP, potassium, and plasma renin. In this case, addition of a medication that specifically blocks aldosterone's effect on renal NaCl reabsorption can be helpful. Aldosterone specifically stimulates the distal epithelial cell Na⁺ transporter, ENaC, which contributes to the volume expansion and resultant hypertension. Use of ENaC inhibitors, such as amiloride or triamterene, in combination with MRBs may be helpful. We typically use amiloride, starting at 5 mg by mouth daily and titrating as needed.

Aldosterone synthase inhibitors are a new treatment option under development.⁷⁰ These medications directly inhibit adrenal aldosterone synthesis, leading to decreased plasma aldosterone levels and improvements in BP control. Because they do not target MR, their side effect profile is likely to differ from that seen with MRBs, although the development of hyperkalemia is not likely to differ.

Considerations in the Treatment of Patients With Chronic Kidney Disease

CKD is common in patients with primary aldosteronism,^{71,72} which leads to several concerns regarding changes in serum/plasma K⁺

and GFR. This frequency may be because of direct effects of aldosterone to stimulate renal fibrosis and glomerular injury, manifested initially as albuminuria, and to indirect effects resulting from the associated hypertension and, if present, hypokalemia. Because aldosterone also stimulates glomerular hyperfiltration, which may contribute to renal injury, the full extent of renal injury may not be evident until the primary aldosteronism is adequately treated. Anticipating and then appropriately responding to these effects is helpful for the management of the patient with CKD and primary aldosteronism.

Patients with CKD who are treated with an MRB may be more likely to develop clinically significant hyperkalemia during treatment. Treatment with a K⁺-binding polymer, such as patiromer or sodium zirconium cyclosilicate, can facilitate treatment, enabling continued MRB use with less risk of hyperkalemia,⁷³ and the beneficial effect is maintained even if eGFR is between 25 and less than 30 mL/min/1.73 m².⁷⁴ Our recommendation is to follow closely patients who have CKD who are started on MRB therapy for the development of hyperkalemia. If it occurs, we then start therapy with a K⁺-binding resin (see above), dosed as needed to control the serum/plasma K⁺. Although this approach is supported by the benefit of MRB treatment to improve BP and to slow the progression of CKD, whether correcting the hyperkalemia by addition of a K⁺ binder versus discontinuation of MRB deserves further study.

Second, initiation of MRB therapy can be associated with an acute decrease in eGFR in patients with CKD. However, patients with a greater acute decrease had a slower long-term eGFR decline.⁷⁵ Acute decreases in eGFR after initiating therapy with an MRB does not need to be an indication to discontinue therapy.

SELF-ASSESSMENT QUESTIONS

- The most common genetic cause of primary aldosteronism because of an APA is related to a mutation in which protein?
 - Aldosterone synthase
 - Mineralocorticoid receptor
 - AT1 receptor
 - Potassium channel (KCNJ5)
 - Sodium channel (ENaC)
- Based on recent comparative studies, which of the following MR antagonists for primary aldosteronism is preferred and for what reason?
 - Eplerenone because of greater efficacy
 - Eplerenone because of fewer side effects and less drug discontinuation
 - Spironolactone because of greater efficacy
 - Spironolactone because of fewer side effects and less drug discontinuation
- Which of the following is the preferred first-line test for identification of primary aldosteronism?
 - Serum potassium
 - Plasma aldosterone
 - Plasma renin activity
 - ARR in combination with absolute level of plasma aldosterone
 - Urinary sodium and potassium excretion rate

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